

Problem

Calculate these complex numbers and express your results in rectangular form:

a. $\frac{60\angle 45^\circ}{7.5 - j10} + j2$

b. $\frac{32\angle -20^\circ}{(6 - j8)(4 + j2)} + \frac{20}{-10 + j24}$

c. $20 + (16\angle -50^\circ)(5 + j12)$

Step-by-step solution

Step 1 of 3

(a)

The complex function is $\frac{60\angle 45^\circ}{7.5 - j10} + j2$.

Convert the complex number into rectangular form.

$$\begin{aligned}\frac{60\angle 45^\circ}{7.5 - j10} + j2 &= \left(\frac{42.426 + j42.426}{7.5 - j10} \right) + j2 \\ &= -0.6788 + j4.752 + j2 \\ &= (-0.6788 + j6.752)\end{aligned}$$

Therefore, the rectangular form of the complex number is $\boxed{-0.6788 + j6.752}$.

Step 2 of 3

(b)

The complex function is $\frac{32\angle -20^\circ}{(6 - j8)(4 + j2)} + \frac{20}{-10 + j24}$.

Convert the complex number into rectangular form.

$$\begin{aligned}\frac{32\angle -20^\circ}{(6 - j8)(4 + j2)} + \frac{20}{-10 + j24} &= \frac{30.07 - j10.945}{40 - j20} + \frac{20}{-10 + j24} \\ &= 0.71085 + j0.0818 - 0.296 - j0.71 \\ &= 0.41485 - j0.6282\end{aligned}$$

Therefore, the rectangular form of the complex number is $\boxed{0.41485 - j0.6282}$.

Step 3 of 3

(c)

The complex function is $20 + (16\angle -50^\circ)(5 + j12)$.

Convert the complex number into rectangular form.

$$\begin{aligned}20 + (16\angle -50^\circ)(5 + j12) &= 20 + (10.285 - j12.257)(5 + j12) \\&= 20 + 198.5 + j62.13 \\&= 218.50 + j62.13\end{aligned}$$

Therefore, the rectangular form of the complex number is $218.50 + j62.13$.